PRE-ALGEBRA (8TH) · UNIT 6: CONGRUENCE AND SIMILARITY

Tile Pattern Studio

Design floor tile patterns with transformations, similarity, and angle facts.

Honors

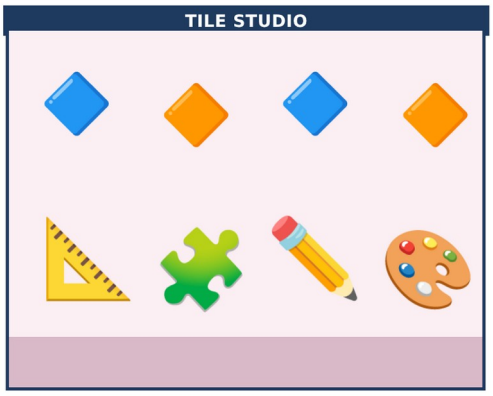
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A community center is redoing its lobby floor and wants a tile pattern designed by students. The pattern starts with one right-triangle tile that is moved around a grid.

You will use transformations, similarity, and angle relationships to build a pattern that fits perfectly and to measure things you cannot reach.



Starting Tile

Triangle ABC

A(1, 1), B(4, 1), C(1, 3)

Rules

Rigid motions keep size and shape

Dilations keep shape

Learning Goals

- I can find single transformations equivalent to sequences and prove it with coordinates.
- I can use dilations about any center and similarity relationships.
- I can use angle facts to decide which polygons tile a floor.

Project Timeline

When	What you will do
Day 1	Equivalent Moves
Day 2	Similarity at Work
Day 3	Which Shapes Tile?
Final	Tile Design Portfolio + reflection

Part 1: Equivalent Moves

Day 1 · Tile ABC: A(1, 1), B(4, 1), C(1, 3).

1. Show that reflecting over the y-axis and then over the x-axis is the same as one rotation. Which one?

2. Find a transformation (or sequence) that maps ABC to A″(−1, −1), B″(−1, −4), C″(−3, −1).

3. Dilate the tile by a scale factor of 2 centered at C(1, 3).

Part 2: Similarity at Work

Day 2 · Use dilations and similar triangles.

4. A tile is dilated by a scale factor of 3. The original area is 3 square units. Find the new area.

5. A student's eyes are 5 feet high. She stands 4 feet from a floor mirror and sees the top of a building 48 feet from the mirror. How tall is the building?

6. Two grout lines are parallel and two transversals meet at a point, forming two triangles. Explain why the triangles are similar.

Part 3: Which Shapes Tile?

Day 3 · A shape tiles a floor at a point if its angles fit exactly around that point (360°).

7. Explain why regular hexagons tile a floor but regular pentagons do not.

8. A triangle has angles (2x)°, (3x + 5)°, and (4x − 14)°. Find each angle.

9. Design a tessellation that uses the original tile and its images under two different transformations. Describe the moves and show why the angles fit.

Final Product: Tile Design Portfolio

Your final product must include:

- Coordinate proofs of equivalent transformations
- Dilation about a point other than the origin
- Indirect measurement with the mirror method
- Tessellation analysis with angle sums
- Designed tessellation with documentation

Reflection

R1. How can you prove two sequences of transformations are equivalent?

R2. Why do any triangle and any quadrilateral tile a floor?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Transformations, dilations, similarity, and angle relationships are applied correctly. Goes beyond the minimum with a creative, realistic choice.	Transformations, dilations, similarity, and angle relationships are applied correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Portfolio shows coordinates, rules, measurements, and a clear final design. Polished, organized, and visually engaging.	Portfolio shows coordinates, rules, measurements, and a clear final design. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.